

Design FMEA — V_K HEV Battery Cell

****Case**:** HEV Battery Design - V_K Extreme Transport

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Failure Mode and Effects Analysis

| Failure Mode | Potential Effect | Potential Cause | Current Design Control | Severity (S) | Occurrence (O) | Detection (D) | RPN | Action

|---|-----|-----|-----|-----|-----|-----|-----|-----|

1 | Lithium plating at 4C | Internal short circuit, capacity fade, safety hazard | Insufficient Li⁺ transport at high rate | High porosity (0.45), thin separator (8μm), small particles (3μm); DFN verified anode_potential > 0V | 9 | 2 | 3 | 54 | Monitor anode potential in production QC

2 | Excessive SEI growth at elevated temperature | Capacity fade, increased impedance | High temperature accelerates SEI formation kinetics | Chen2020 SEI model verified at 45°C: 536 nm < 550 nm limit | 6 | 3 | 4 | 72 | Consider FEC additive for wider margin

3 | Thermal runaway on nail penetration | Cell fire, propagation to adjacent cells | Internal short from nail, exothermic reactions | Verified: no TR at 10W heat input (hA=0.5); cell mass provides thermal buffer | 10 | 1 | 3 | 30 | Maintain separator integrity in production

4 | Overcharge-induced thermal event | Cell swelling, venting, potential fire | Charging above voltage cutoff | Verified: T_{max}=34°C at +0.5V overcharge; BMS voltage cutoff required | 9 | 1 | 2 | 18 | BMS overvoltage protection mandatory

5 | High porosity reduces mechanical strength | Electrode delamination, particle disconnection | Porosity 0.45 exceeds standard 0.33 | Not tested in simulation (outside scope); production validation needed | 7 | 3 | 5 | 105 | Mechanical testing recommended

6 | Thin separator (8μm) reduces puncture resistance | Internal short from dendrite or manufacturing defect | Separator thinner than standard 25μm | Ceramic coating recommended; not modeled in simulation | 9 | 2 | 4 | 72 | Ceramic-coated separator required

7 | Numerical instability in thermal runaway model | False positive safety assessment | ODE solver instability at low hA | Sensitivity analysis shows hA>0.1 gives stable results | 5 | 2 | 2 | 20 | Use hA≥0.5 for nail penetration modeling

****RPN Scale**:** Severity (1-10) × Occurrence (1-10) × Detection (1-10). RPN > 100 requires action.

****Highest RPN items**:** #5 (RPN=105, mechanical strength at high porosity) — requires physical testing validation.

****Design risk summary**:** The V_K design achieves all simulated performance targets but introduces manufacturing risks from high porosity and thin separator. These are identified as areas requiring physical prototype validation.